Furthose samreen B

CO1-AT1-ASSESSMENT TOOL

1. Water Jug Problem (4-gallon and 3-gallon jugs)

Goal: Get exactly 2 gallons in the 4-gallon jug.

Step

4-Gallon Jug

3-Gallon Jug

1. Fill 3-gallon jug

0

3

2. Pour into 4-gallon jug

3

0

3. Fill 3-gallon jug again

3

3

4. Pour into 4-gallon until full

4

2

5. Empty 4-gallon jug

0

2

6. Pour remaining 2 gallons into 4-gallon jug

2

0

Answer: The 4-gallon jug contains exactly 2 gallons.

2. Mars Rover Agent

i. Percepts

Camera images

Rock and soil samples

Temperature

Distance to obstacles

Battery level

ii. Operating Environment

Partially observable

Dynamic

Sequential

Unknown

Real-world environment

iii. Actions

Move forward/backward

Turn left/right

Capture images

Collect soil/rock samples

Analyze samples

Send data to Earth

iv. Performance Measure

Correct sample collection

Safe navigation

Minimum energy usage

Successful data transmission

Mission completion

v. Suitable Agent Architecture

Learning Agent

Reason: It learns from experience and adapts to the unknown Martian environment.

3. 8-Queens Problem

Problem Components

Initial State

Empty 8×8 chessboard.

State

Queens placed on the board.

Actions

Place one queen in a safe position.

Goal Test

Place all 8 queens so that no two queens attack each other.

Path Cost

Each move costs 1.

Search Strategy

Backtracking Search

Depth-First Search (DFS)

4. OLA Cab Booking

Problem-Solving Agent

Goal-Based Agent

Reason: It selects the best cab to reach the destination.

Pseudocode

START

Input pickup and destination

Display available cabs

User selects cab type

Calculate fare and travel time

Confirm booking

Assign nearest driver

Driver picks up customer

Reach destination

END

5. Uniform Cost Search (UCS)

Algorithm

Start from node S.

Add S to the priority queue.

Remove the node with the lowest cost.

Expand its neighboring nodes.

Update the path cost.

Repeat until destination G is reached.

Return the least-cost path.

Result

Least-cost path: S → … → G (choose the path with the minimum total cost).

Algorithm Used: Uniform Cost Search (UCS).

Advantage: Always finds the optimal (minimum-cost) path.